A Value-Based Strategic Management Process for e-Government Strategy Planning and Performance Control

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ABSTRACT
Creating values for citizens, businesses, as well as governments is the mission for launching e-government strategies and services. The effectiveness of e-government strategies depends heavily on the cause-and-effect relationships among strategic objectives, action plans, as well as performance outcomes identified and specified during the strategy formulation, project implementation, and performance measurement processes. The goal of this paper is to propose a value-based strategic management framework and process for supporting efficient and effective planning, implementation, as well as evaluation of e-government strategies. In the proposed process, concepts and methodologies including the balanced scorecard, strategy map, and strategy gap are adapted and integrated to illustrate the constructs, activities, and procedures of e-government strategic management.

Categories and Subject Descriptors
K.6.1 [**Management of Computing and Information Systems**]: Project and People Management – *Life cycle, Management techniques, Strategic information systems planning.*

General Terms
Management, Measurement, Performance, Design, Economics

Keywords
Value Creation, Strategic Management, E-Government Strategies, Balanced Scorecard, Strategy Map, Strategy Gap

1. INTRODUCTION
To take advantages of web technologies as what businesses have done and gained a great success in delivering online services to their customers, governments in many countries have directed e-government (EG) strategies to foster the movement towards public service digitalization. Powerful e-government systems and portals enable citizens, businesses, and government agencies to access more desired government information and integrated services in a more convenient way [4][17][22][41][44]. Creating values for stakeholders including citizens, businesses, as well as government agencies is regarded as the mission for launching e-government strategies and services.

However, how to identify and assess potential values is still a less-touched issue in the literature [15][28]. In addition, the effectiveness of e-government strategies depends heavily on the cause-and-effect relationships among strategic objectives, action plans, and performance outcomes generated during the strategy formulation, plan implementation, and performance evaluation processes. The gaps between the projected and actual performance outcomes with respect to strategy planning and implementation are also needed to be controlled. Without properly handling these strategic management issues, it would be difficult for governments to leverage organizational capabilities as well as information society readiness in order for fully achieving the objectives of e-government strategies.

Although the literature has reported a great deal of e-government initiatives [6][21][25][29], a variety of shortcomings related to strategic management still exist in previous works. Firstly, the scopes of analyzing and measuring values are too narrow to take into account all e-government participants' needs. Secondly, there is no clear guideline for specifying and verifying relationships between various e-government strategies, as well as among values, visions, missions, strategic objectives, action plans, and performance measures. Ambiguous strategies with unclear strategy-action-performance links will definitely create barriers for reaching the anticipated performance outcomes. Thirdly, there is no effective strategic management method for directing the e-government strategy formulation, implementation, and performance evaluation processes. The significant lack of appropriate strategic management framework and process in the public sector hinders government agencies from successfully carrying out the strategy-related thinking, planning, analysis, design, implementation, execution, communication, evaluation, and control activities. Aiming at filling the literature gap, this paper proposes a value-based strategic management framework and process for efficiently and effectively formulating, implementing, and evaluating e-government strategies. In our value-based strategic management approach, various concepts and methodologies including the balanced scorecard (BSC), strategy map (SM), and strategy gap (SG) are adapted and integrated to systematically drive the processes of specifying strategic perspectives, missions, objectives, and performance measures, as well as identifying strategic gaps for better managing e-government strategies.

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The rest of this paper is organized as follows. In section 2, brief reviews on the concepts and processes of value proposition and strategic management, as well as the BSC, SM, and SG methods are provided. Section 3 outlines and describes the steps for developing EG-related strategy maps based on the value-centric BSC perspectives, as well as identifying strategic gaps within and across different perspectives in the established strategy maps. In section 4, a case of EG strategies in Taiwan is presented to show the feasibility, efficiency, and effectiveness of using the proposed value-based strategic management process to manage EG strategies in a specific country. The final section contains summary and concluding remarks.

2. LITERATURE REVIEW

In this section, a brief review on the value proposition, strategic management process, balanced scorecard, strategy map, and strategy gap methods for developing and controlling business and/or public strategies is provided.

2.1 Values and Strategic Management

In the e-business literature, value has been noted as an integral element of business models, and in the meantime taking the central role in strategic planning [42]. Many researchers also argue that as a prerequisite to strategic management, business performance measurement (BPM) should focus on value proposition and value creation [8][10][12]. In the business domain, issues related to value identification and value assessment have often been addressed using different perspectives. Value class that has been brought to attention include customer value, shareholder value, business value, relationship value, product and service value, as well as values of information technology and innovation. For example, Ulaga (2003) presents eight value dimensions, namely, product quality, service support, delivery, supplier know-how, time-to-market, personal interaction, direct product costs, and process costs, to highlight manufacturer-supplier relationships [39]. On the other hand, among researches that discuss strategic management process (SMP), Coveney et al. (2003) consider SMP as an essential part of the corporate performance management (CPM) [5]. They think a SMP should integrate strategies, tactics, and metrics, and encompass three sub-processes including strategic analysis, strategy formulation, as well as strategy implementation. They also group mission, objectives, goals, strategies, tactics, and key performance indicators as components of a strategic plan. Other research efforts in this area focus their concerns mainly on strategic thinking, strategic planning, or strategy gap analysis that respectively put more weights on addressing vision and mission, objectives and strategies, or differences between goals and executions [11][13][30][31].

In the e-government literature, research works addressing issues regarding value proposition, strategic management, and performance measurement, either independently or integratively, are very limited. Although getting organizational, institutional and/or stakeholder values has long been considered fundamental to underpin government strategies, no detail discussion concentrating on value proposition and assessment has been provided [15][28]. Values previously described in the government context mainly focus on performance of public activities and services that are often characterized by quality, efficiency, and effectiveness. For instances, Montagna (2005) presents a 5-dimension with 4-criteria framework for the assessment of e-government proposals, eventually e-services projects [32]. Five dimensions of the framework consist of product, time, distance, interaction and procedures, while each dimension can be evaluated by four criteria including efficiency, effectiveness, strategic benefits, as well as transparency and institutional value. Papadomichelaki et al (2006) organize main influential quality-related components for e-government services into four key areas: service, content, system and organization [34]. In each key area, quality elements are identified. For an example, accuracy, consistency, in time, interaction, trust, and degree of personalization are chosen as major quality concerns of the service key area. Charalabidis et al. (2006), on the other hand, indicate the lag of a complete taxonomy for organizing municipal e-services to meet the need of citizens and businesses [4]. They present 7 main municipality e-services categories with 11 classification facets for describing service characteristics. Besides, emphasizing on the delivery of customer-centric public services, King (2007) identifies 3 new relationships between local authorities and citizens including informational/transactional, council-driven, and community-support for system design considerations [22]. Summarizing different viewpoints from the previous research, it can be seen that for e-government projects, customer (citizens and/or businesses) values and organization values are created through the creation and delivery of e-service values.

As for concerning with strategic management and performance measurement issues, Wimmer (2002) proposes a layered framework in which vision, strategies, initiatives, projects, and applications are organized in a hierarchy for developing e-government applications [41]. In a survey of e-government practices in the leading countries, Lee, Tan, and Trimi (2005) report the paths of strategic planning and implementation of EG initiatives in these countries [25]. For instances, in the U.S., and started from 2002, the EG strategy was planned and then an action plan including 25 EG initiatives were implemented. Furthermore, results of the EG adoption at the state level were examined by means of a set of distinct sectors including social services, digital democracy, e-commerce, taxation, and revenue. In member countries of the European Union (EU), supporting EG was set as one of the primary goals of the eEurope initiative launched by the European Commission (EC) in 1999, and in the subsequent eEurope 2002 and 2005 Action Plans, detailed description of actions and actors were provided. However, specific action plans for EG initiatives had not been established, and EG as a whole was put among the 23 key indicators of the e-Europe benchmarking program to assess differences among the EU member nations in the adoption of emerging Internet related technologies.

Integrating and extending the reviewed SMP concepts to cover the needs of both the e-business and e-government sectors, an effective strategic management process should involve the undertakings of strategic thinking and analysis, strategic planning and strategy formulation, strategy implementation and action plan execution, as well as performance evaluation and control. The strategic thinking, planning, and formulation phases focus on identifying values, specifying the current status and future goals of value proposition and delivery, and proposing feasible paths that lead to the desired ends of value creation. At first, values, visions, and missions are identified, and strategic issues, critical success factors, as well as potential solutions are explored. Next, strategic objectives are specified, specific strategies are formulated, and required resources are allocated for those formulated strategies. In the strategy implementation, execution, and performance evaluation phases, initiatives, action plans, and performance measures associated with strategies and their strategic objectives are developed, then, action plans are launched and their performance outcomes are measured. During the strategy formulation and the performance measurement processes, strategic gaps with respect to planning and performance results are identified and controlled. Basically, a strategy is considered effective when goal-oriented action plans are executed and specified strategic objectives are reached with high satisfaction levels reflected in actual outcomes of performance indicators.

2.2 The Balanced Scorecard

The balanced scorecard, developed by Kaplan and Norton (1996), is basically a management instrument for measuring business performance from four balanced perspectives, namely, financial, customer, internal process, and learning and growth [18]. It is further deemed as a strategic management tool for translating strategies into values [19]. The BSC is commonly characterized by its capability to combine management concerns from financial and non-financial, internal and external, as well as current and future perspectives, and is noted as the most adopted concept and tool in the field of business performance measurement [27].

Although the concept and business applications of the BSC have been extensively reported, research on the adoption of the BSC in the public sector is still in its early stage. Griffiths (2003), in examining the BSC adoption in three New Zealand public sector organizations, report that none of the case organizations has implemented a BSC fully because causal linkages between the measures in the four BSC perspectives have not been established [9]. Chan (2004), in his survey on using the BSC in municipal governments of the USA and Canada, note that there is only limited use of the BSC for performance measurement [3]. He conclude that although most governments have developed measures in various aspects to assess organizational performances including financial, customer satisfaction, operating efficiency, innovation and change, and employee performance etc., more studies are needed to address issues such as how to modify and integrate existing measures with the BSC perspectives, as well as what are the role and benefits of the BSC for performance measurement and strategic management in public sectors. Yu and Wang (2005), in their attempt to use the BSC for managing e-government related digital divide (DD) strategies, propose a DD-BSC with four aligned perspectives including beneficiary, governmental functions and processes, nation-wide learning and growth, and financial to better fit the broad spectrum of DD factors and indicators [43]. From reviewing the EG related BSC literature, the need to establish causal links between values, strategies, and performance measures within the BSC context is significant and essential for making the BSC an effective strategic management tool in EG development, and thus the issues of adopting and adapting the BSC to the EG domain deserve further exploration.

2.3 The Strategy Map

The strategy map, developed and used to depict cause-and-effect relationships between performance drivers and desired outcomes, enables an organization to illustrate clearly its objectives, initiatives, targets, performance measures, and their linkages [19]. Kaplan and Norton (2003) suggest that the best way to develop strategy maps based on the BSC is from the top down, starting with the destination (ends) and tracing back the routes (means) that lead to it [20]. In other words, for achieving the final goal of creating financial values (the top-level financial perspective), a company need to deliver values to their customers (the second-level customer perspective) through providing process efficiency (the third-level internal process perspective), and in order to do that, they need to build up organizational capabilities through learning and innovation (the bottom-level learning and growth perspective). However, they also indicate that, for nonprofit and government units, although the top level view has often been replaced by customers or constituents instead of using financials, the systematic approach for developing strategy maps and the illustration of links between strategic objectives, actions, and performances remain as major challenges to all public policy makers and researchers. In an experience report of adopting the strategy map approach to develop strategies in the UK Small Business Service, Irwin (2002) creates a strategy map based on a modified BSC with four top-down layers including Fulfilling customer needs, Understanding our customers, Improving our performance, and Finance & organization perspectives respectively [15]. In the strategy map, organizational objectives are arranged into the 4 BSC levels and logical links are set to illustrate the relationships between these objectives. It is concluded that the real strength of the strategy map is that, as a communication tool, it enables people to see how their particular functions within the organization fitted into the whole, and consequently, helps to explain the strategy to all parties and to overcome internal divisions. Moreover, strategy maps can also help an organization to detect major gaps in the strategies being implemented at lower organizational levels. As a summary, an unified model and a systematic process for creating a BSC-based strategy map suitable for use in the EG sector remain as critical issues to be tackled.

2.4 The Strategy Gap

The strategy gaps are viewed as the results of missing links between strategies, strategic objectives and performance measures. In an earlier work of conceptualizing strategic gap, Harrison (1989) indicates that different names and meanings have been given previously for describing strategic gaps such as the planning gap, uncertainty gap, performance gap, positive/negative gap, and total gap etc. [11]. He instead, defines the strategic gap as an imbalance between organization's current actual position and future desired position. He also indicates that the strategic gap can be measured by assessing and comparing internal organization capabilities and external environment conditions. The strengths and weaknesses of organizational capabilities are reflected in management, technology, policies, and resources aspects, while the impact factors of the external environment include opportunities, threats, requirements, and responsibilities. On the other hand, Proctor (1997), when dealing with the marketing strategies, detects strategic gaps by comparing the sales performance with the required objectives based on internal factors including financial, process, and product [35]. Focusing on corporate performance management issues, Coveney et al. (2003) group the gaps between strategy and execution into three areas, namely, the management-induced, the process-induced, and the technology system-induced gaps [5]. The causes of management-induced gaps include failure to secure plan support, failure to communicate the strategy, failure to adhere to the plan, and failure to adapt to significant changes. The set of process-induced gap causes includes lack of strategic focus, calendar based, financially focused, internally focused, and lack of realistic forecasting. Issues related to the technology system-induced gaps include fragmented systems and misplaced dependence on enterprise resource planning (ERP). They further indicate that what causes the gaps and what can be done to close it remain as crucial questions to be answered. Kaplan and Norton (2003), taking a view from the strategy map, consider strategy gaps as the missing supports of the lower level strategies to the upper level strategies [20]. For instance, a strategy gap exists when no strategic objectives or metrics are specified to establish dealer relations in the customer level in order for supporting the revenue growth strategy in the financial level. In the public sector, however, there is no previous research work addressing the issue of strategy gaps. How to define, identify, detect, and control strategy gaps when formulate and evaluate EG strategies, as well as how to build links connecting strategy gaps with the BSC and strategy maps for EG strategic management are still major issues calling for more in-depth studies.

As can be seen in the literature, although the issues of strategy framework and causal relationships for structurally linking e-government values, visions, strategic objectives, action plans, and performance measures have been considered critical, not much research had been done in this areas. Furthermore, previous research efforts regarding the adaptation and integration of the BSC, SM, and SG methodologies for managing and evaluating e-government strategies are still very rare, if not totally absent. Therefore, there is a great research potential and practical importance to explore the issues of developing a unified strategic management framework, and integrating the BSC, SM, and SG approaches to facilitate the entire strategic management process of e-government strategies.

3. THE VALUE-BASED SM PROCESS

In this section, systematic processes for developing the value-based BSC and BSC-based strategy maps, as well as for identifying strategy gaps within the strategy maps for efficiently and effectively managing EG strategies are illustrated.

3.1 The Value-Based Strategic Process

The complete value-based process for EG strategic management is outlined as follows and illustrated in Figure 1.

1) Identifying EG values: EG related values to be proposed to and created for all stakeholders including citizens, businesses, local and municipal government agencies, as well as the society and the whole nation are identified. Also specified are value dimensions and associated metrics.
2) Identifying visions and missions: The visions set the anticipated future outcomes and positions associated with the value creation. The missions state what has to be accomplished to reach the desired future positions.
3) Specifying strategies, strategic objectives and performance indicators: In order to undertake missions, strategies are formed and strategic objectives are specified. Key performance indicators (KPIs) that measure the achievement of strategic objectives are also generated and classified.
4) Defining BSC perspectives and constructing the value-based BSC: Once the value dimensions are classified, they can be grouped and used to define four BSC perspectives. And then within each of the four BSC perspectives, missions, strategies, strategic objectives, and performance indicators are properly organized and presented.

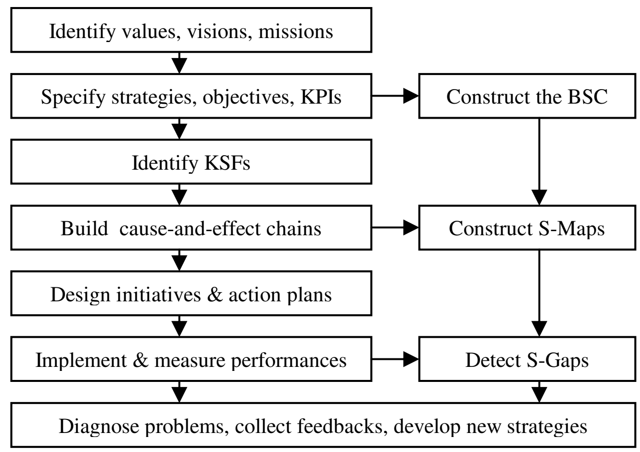

Figure 1. The value-based strategic management process

5) Identifying key success factors (KSFs) and key impediments: For successfully fulfilling the strategic objectives, critical success factors and potential impediments are identified to guide the design of suitable actions for securing the goals.
6) Building cause-and-effect chains: Taking strategic objectives and key success factors under consideration, all possible activities towards achieving the strategic objectives are explored and then a cause-and-effect chain is built to illustrate structural relationships among strategies, strategic objectives, and performance indicators, as well as ends-and-means links within and across the BSC perspectives.
7) Constructing BSC-based strategy maps: Adopting a layered approach, the four BSC perspectives are rearranged and placed into top-down levels. The strategies, strategic objectives, performance indicators and cause-and-effect links are visually presented to facilitate communication, implementation and control of strategies.
8) Detecting planning gaps and refining strategy maps to fill the gaps: Strategic gaps in the strategy planning and formulation stage are detected if there exist missing strategies, strategic objectives, or performance indicators, or missing logical links between strategies, objectives, and measures of the same SM level or across different SM levels, The success of top level strategies are supported by the success of lower level strategies.
9) Designing initiatives and action plans with specified performance targets: For successfully implementing EG strategies, strategy related operable initiatives and action plans are developed, and in the meantime, targeted levels of performance outcomes are specified.
10) Launching action plans as well as assessing and collecting performance outcomes: Action plans are deployed by responsible government agencies with the aims at achieving projected performance outcomes. Once implemented, data of

performance indicators are collected and analyzed to reflect actual performance outcomes.

11) Detecting performance gaps and diagnosing problems: Significant differences between projected and actual performance outcomes indicate the existence of performance gaps. The gap problems are then diagnosed with necessary remedy actions proposed.
12) Deriving new strategies and action plans for improving performance outcomes: To close the performance gaps, as well as to take into account feedbacks collected from processes of EG implementation, operation, and applications, new strategies and/or action plans are derived and implemented for ensuring targeted performance outcomes as well as for reaching the vision states.

3.2 The Value-Based Balanced Scorecard

EG values identified include service values, citizen values, business values, government employee values, organizational values, institutional values, administration values, society values and nation values. These values are classified into five dimensions, namely, services, public users, government agencies and processes, government service chain, and national and global environment. Value metrics for the services dimension consist of quality (consistency, accuracy, timely, innovation, etc.), efficiency (24x7, location, process, interaction, transparency, etc.), effectiveness (personalization, collaboration, satisfaction, benefits, cost savings, etc.), and trust (confidence, support, etc.).

Value-corresponding visions include, for instances, “all public users can easily access high quality EG services”, “organizational capabilities of local and municipal government agencies are strong in innovating and delivering EG services”, “there is an one-stop EG service portal offering 24x7 and transparent EG services”, and “the levels of information society readiness and national competitiveness are high and digital divides are low”, etc. The missions are then clarified as to create the values for public users, government agencies, government service chain, as well as society and nation.

According to the identified values, visions, and missions, the BSC perspectives can be defined as Public Beneficiaries, Government Structure and Process, Government Service Chain, as well as National and Global Environment. Based on the BSC perspectives, specified strategies include public user strategies, internal government strategies, government service chain strategies, as well as national ICT, learning, growth, and competition strategies.

These perspective-level strategies can then be further decomposed into sets of related sub-strategies, for instances, citizen strategy and business strategy for the public beneficiary perspective, as well as national ICT strategy, national learning strategy, digital divide strategy, and global competition strategy for the national and global environment perspective.

For the Public Beneficiaries perspective and the public user strategies, strategic objectives include: to provide easy access to EG systems and services, to deliver citizen and business desired information and services, to enforce personalization and customization in EG services, to provide more community, collaborative, and participative services, to enhance public (citizen/business) relationship management, to provide a security and trust EG service environment. Public associated performance indicators may include number of registered public users (citizens/businesses), cost and time for requesting and receiving services, EG maturity and usage indexes, levels of personalization, collaboration, and participation, levels of public (citizens/businesses) satisfaction on quality, efficiency, effectiveness, trust, and completeness of EG services.

For the internal government strategies with respect to the Government Structure and Process perspective, strategic objectives include: to set up employee communication, collaboration and learning channels, to leverage organizational capabilities on service development, operation and knowledge management, to improve internal operational processes, to increase organizational productivity, to reduce operational costs, to achieve service/decision effectiveness, to create organizational image and awareness, to develop and deliver innovative and value-added services. And accordingly, the performance indicators include HR skill levels and productivity ratios, level of employee satisfaction, level of organizational capabilities, levels of Internal operating efficiency and effectiveness, level of cost savings, levels of service innovation and utilization, levels of public satisfaction on agencies (local and municipal) and their services.

For the Government Service Chain perspective and related strategies, strategic objectives include: to establish vertical and horizontal government service chains, to enhance service chain management and operations, to develop information and value sharing policies, to facilitate inter-organizational service integration and delivery processes. Performance indicators associated with these service chain objectives include quality and response time to user-requested integrated services, cost and time reductions in inter-organizational information and transaction processing, levels of inter-organizational services integration and operational transparency, levels of benefit generation and cost reduction for service chain participants and the entire government service chain, levels of public satisfaction on time/location conveniences and output qualities of inter-organizational services.

For the national learning, growth, and competition strategies related to the National and Global Environment perspective, strategic objectives specified include: to establish EG related national information infrastructure and one-stop service portal, to set up a nation-wide e-learning environment, to close digital divides, to sustain national economic growth, to leverage national capability, to achieve high level information society readiness, to sustain high level world competitiveness, etc. Associated performance indicators include digital divide status, information society index, world competitiveness index, levels of completeness, easiness, and security of one-stop EG service portal, levels of national learning, innovation, and growth, levels of budget efficiency and effectiveness, and return on investment (ROI), etc.

Up to this point, a value-based BSC can be established by arranging and placing value-based missions, strategies, strategic objectives, and performance indicators into these four BSC perspectives. Focusing on EG services, Figure 2 depicts a service-centered value framework.

Table 1 illustrates a corresponding perspective value-based BSC with specified missions, strategies, strategic objectives, and performance indicators.

Table 1. A value-based BSC

National/Global Env. Perspective	Government Service Chain Perspective
Mission: To create society and nation values. **Objectives:** To establish EG related national information infrastructure, To provide an one-stop service portal, To set up a nation-wide e-learning environment, To close digital divides, To sustain national economic growth, To leverage national capability, To achieve high level information society readiness, To sustain high level world competitiveness, etc. **Performance indicators** Digital divide status, Information society index, World competitiveness index, Levels of completeness, easiness, and security of one-stop EG service portal, Levels of national learning, innovation, and growth, Levels of budget efficiency and effectiveness, Return on investment (ROI), etc.	**Mission:** To create service supply chain values. **Objectives:** To establish vertical and horizontal government service chains, To enhance service chain management and operations, To develop information and value sharing policies, To facilitate inter-organizational service integration and delivery processes, etc. **Performance indicators** Quality and response time to user-requested integrated services, Cost and time reductions in inter-organizational information and transaction processing, Levels of inter-organizational services integration and operational transparency, Levels of benefit generation and cost reduction for service chain participants and the entire government service chain, Levels of public satisfaction on time/location conveniences and output qualities of inter-organizational services, etc.
Gov. Structure and Process Perspective	Public Beneficiaries Perspective
Mission: To create employee and organization values. **Objectives:** To set up employee communication, collaboration and learning channels, To leverage organizational capabilities on service development, operation and knowledge management, To improve internal operational processes, To increase organizational productivity, To reduce operational costs, To achieve service/decision effectiveness, To create organizational image and awareness, To develop and deliver innovative and value-added services, etc. **Performance indicators** HR skill levels and productivity ratios, Level of employee satisfaction, Level of organizational capabilities, Levels of Internal operating efficiency and effectiveness, Level of cost savings, Levels of service innovation and utilization, Levels of public satisfaction on agencies (local and municipal) and their services, etc.	**Mission:** To create citizen and business values. **Objectives:** To provide easy access to EG systems and services, To deliver citizen/business desired info. and services, To enforce personalization and customization in EG services, To provide more community, collaborative, and participative services, To enhance public (citizen/business) relationship management, To provide a security and trust EG service environment, …, etc. **Performance indicators** Number of registered public users (citizens/businesses), Cost and time for requesting and receiving services, EG maturity and usage indexes, Levels of personalization, collaboration, and participation, Levels of public (citizens/businesses) satisfaction on efficiency, effectiveness, trust, and completeness of EG services, ..etc.

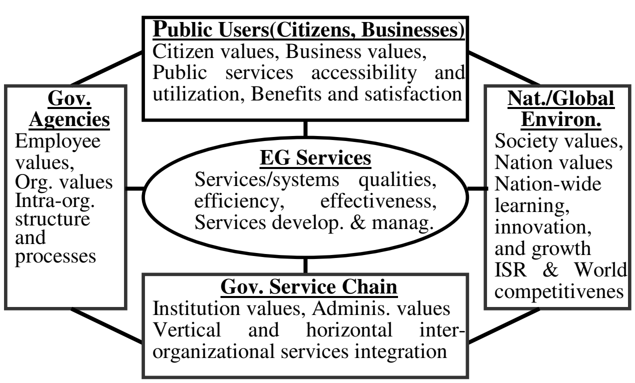

Figure 2. A service-centered value framework for EG

3.3 The BSC-Based Strategy Maps

For strategic objectives presented in the BSC perspectives, critical success factors and key impediments are identified to aid in achieving success and simultaneously avoiding obstacles of the strategy implementation. Examples of key success factors include meeting user needs, providing usable and easy-to-use services and systems, guaranteeing high levels of security, privacy, and trust, managing substantial funding and resources, reengineering government processes, increasing middle management participation, providing organizational training and development environment, managing employee relationships and organizational knowledge, coordinating cross-border public services and activities, managing supply chain of government services, avoiding digital divides, sustaining information society readiness and world competitiveness, adopting holistic strategic management process and EG system development approach, applying useful information technologies, establishing legal environment and national information infrastructure, taking into account multiple perspectives, maintaining top management commitment and high level administrative support, clarifying visions, objectives, and measures, tracking progresses of strategy implementation and operation, enticing user participation, collecting user feedbacks, and managing public relationships, etc. Cause-and-effect chains are then illustrated to show ends and means links that connect strategic objectives with performance outcomes. For example, a cause-and-effect chain specifically links the "to deliver citizen and business desired information and services" strategic objective of the public user strategy in the Public Beneficiaries perspective to its performance outcomes is recorded as the following.

BSC-Perspective (Public Beneficiaries):

- o Strategy(public user)
- o Strategic-Objective(to deliver citizen and business desired information and services)
- o KSF-in-action(meeting citizen and business needs, providing usable services and systems, guaranteeing high levels of security, privacy, and trust, adopting holistic EG system development approach, taking into account citizen and business perspectives, enticing citizen and business participation, collecting citizen and business feedbacks, and managing citizen and business relationships)
- o KPI(cost and time for requesting and receiving services, EG system maturity and usage indexes, levels of citizen and business satisfaction on quality, efficiency, effectiveness, trust, and completeness of EG services).

Consequently, a series of cause-and effect links can be generated for all strategic objectives in all the BSC perspectives. It is possible that different strategic objectives share one or several common KSFs and are associated with partially overlapped performance indicators. The BSC-based strategy maps can then be constructed by firstly placing the four BSC perspectives, i.e. the public beneficiaries perspective, the government structure and process perspective, the government service chain perspective, as well as the national and global environment perspective in a top-down manner, and secondly presenting strategies, strategic objectives, and performance indicators, and their cause-and-effect links within and across associated BSC perspectives. Figure 3 shows a simple example of a BSC-based strategy map.

Based on the complete and correct strategy maps, initiatives are developed to concretize and operationalize strategies, and furthermore, relevant initiatives are packed to form action plans.

3.4 The SM-Based Strategy Gaps

There are basically two types of strategic gaps to be detected, namely, the planning gap and the performance gap. The planning gaps indicate that there exist (1) ambiguous visions, missions, or objectives, (2) misaligned missions and/or objectives, (3) unrealistic objectives and/or measures, (4) missing strategies, strategic objectives, or performance indicators, or (5) missing links between strategies, objectives, and measures within or across SM levels. For example, if "to enforce personalization and customization in EG services" is not specified as a strategic objective in the Public Beneficiaries perspective, then the objective "to enhance public (citizen/business) relationship management" in the same perspective has no sufficient support. Besides, if the objectives in the bottom-level National and Global Environment perspective "to establish EG related national information infrastructure and to provide an one-stop service portal" are not included, then the objective "to facilitate inter-organizational service integration and delivery processes" in the second lower level Government Service Chain perspective, and then the objective "to achieve service/decision effectiveness" in the upper level Government Structure and Process perspective, and consequently the objective "to deliver citizen and business desired information and services" in the top level Public Beneficiaries perspective can not be successfully achieved. The performance gaps, on the other hand, reveal significant differences between targeted and actual performance outcomes when strategy-related action plans have been implemented and operated. For instance, if the global rank of Information Society Readiness was set to be the fifth in the world, but the actual rank turned out to be the tenth, then a performance gap is signified. It is not hard to understand that performance gaps may be caused by planning gaps. Eventually, both the planning and performance gap problems need to be tackled and solved in order to assure the success of EG value creation as well as strategic management. In Figure 3, the dotted boxes or lines, and numbers/words with bracket shown in the strategy map indicate either planning or performance gaps.

4. A CASE EXAMPLE

According to the progress report released by the Research, Development, and Evaluation Commission (RDEC) in Taiwan, steps of the e-government development are outlined as identifying EG driving forces, presenting a roadmap for EG actions, identifying key EG initiatives, establishing a multi-year EG program, setting up performance targets of the EG program, reviewing EG progress status, planning policies and strategies for EG initiative implementation, as well as describing and sharing experiences and lessons learned [36]. In the government directed Electronic Government Program (2001-2004), strategies mentioned include investing for information economy, strategic investment and focus on key applications, public/private partnerships, and marketing e-government. Strategic objectives specified include: to provide online services to all agencies and civil servants, to encourage the government work force at all organizational levels to conduct administrative business and provide public service more efficiently via the Internet, to promote communication and document interchange between organizations, to improve the convenience and efficiency of government services and extend the spatial and temporal coverage of government services by providing 1500 Internet-based application service and one-stop processing services. To locate planning and performance gaps, we first regroup these strategies and objectives into 4 BSC perspectives, place them in a top down manner with objective-related KPIs and associated expected/actual outcomes being presented, and then spot missing items and missing links within and across different perspectives. Table 2 lists a simple case of collected strategies, objectives, KPIs, outcomes, and planning/performance gaps (denoted as G1/G2).

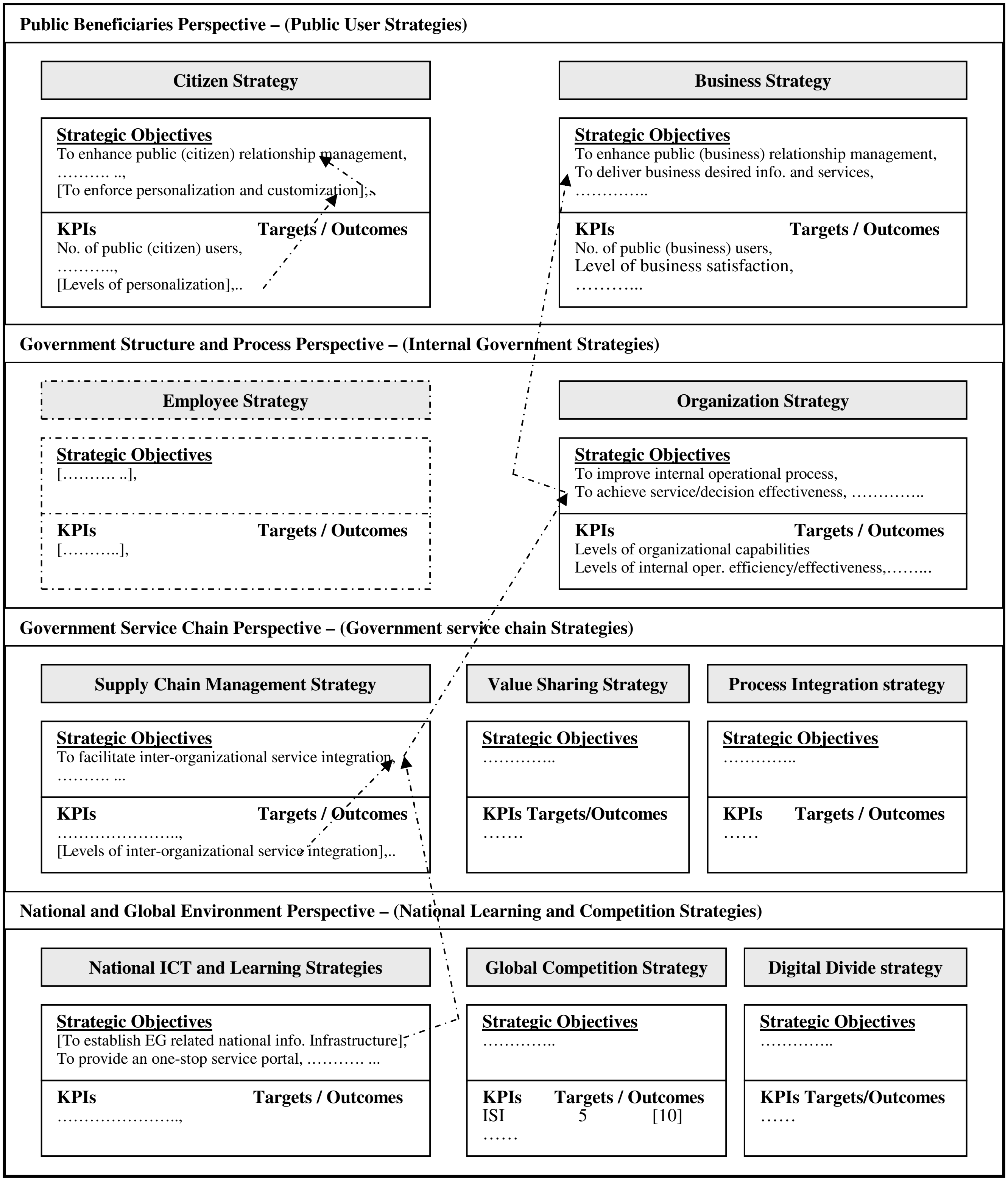

Figure 3. A BSC-based Strategy Map

As we apply the proposed value-based strategic management process to EG projects launched in Taiwan, including general EG strategies and specific digital divide strategies, several facts revealed the need of future improvement on EG related strategic formulation, implementation, as well as performance evaluation activities. The existence of planning and performance gaps include (1) no conceptual model adopted for value proposition, strategic management, as well as performance measurement, (2) some ambiguous missions and objectives, (3) missing key objectives and measures, (4) missing cause-and-effect links between strategies, strategic objectives, and performance indicators, (5) unfulfilled targets, and (6) missing strategic links between multi-year EG projects, etc. A comprehensive strategic management process is strongly demanded for guiding the path towards continuous improvement.

5. CONCLUSION

In this paper we propose a value-based strategic management framework and process for supporting efficient and effective planning, implementation, as well as evaluation of e-government strategies. Value types and dimensions are identified, and value-based BSC, BSC-based strategy maps, as well as SM-based strategy gaps are generated to facilitate the complete EG strategic management process. This comprehensive and systematic EG strategic management approach is feasible and flexible to be adopted to various levels of governments for improving the processes and outcomes of the EG development and deployment. Future studies will focus on detailed discussions regarding strategy map modeling and strategy gap analysis processes, as well as on applying the proposed value-based strategic management approach to specific countries to explore the strength and weakness of using this process.

Table 2. A BSC-SM-SG list for Taiwan EG Projects

Public Beneficiaries perspective		
Strategies: EG marketing strategy, Application strategy, [..],...,[..]	 G1 G1	
Objectives: To improve the convenience and efficiency of government services, To extend the spatial and temporal coverage of government services by providing Internet-based application service, […],..[…].	G1 G1	
KPIs-Target/Actual Outcomes: Internet application services-1500 /[..], [..]-[..]/[..],…..	 G1	 G2 G2
Government Structure and Process perspective		
Strategies: [...],..,[...]	G1	
Objectives: To provide online services to all agencies and civil servants, To encourage the government work force at all organizational levels to conduct e-administrative business and provide efficient public e-service, […], ..,[...].	G1 G1 G1	
KPIs-Target/Actual Outcomes: [..]-[..]/[..],…..	G1	G2
Government Service Chain perspective		
Strategies: Public/private partnerships strategy, […],..,[…]	 G1 G1	
Objectives: To promote communication and document interchange between organizations, […],..[…].	G1 G1	
KPIs-Target/Actual Outcomes: [..]-[..]/[..],…..	G1	G2
National and Global Environment perspective		
Strategies: Investment strategies (information economy, key applications), […],..,[…]	G1 G1	
Objectives: To provide one-stop processing services, […],..[…].	G1 G1	
KPIs-Target/Actual Outcomes: [..]-[..]/[..],…..	G1	G2

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